Rotary Wing Aerodynamics (AE 667)



Wind Turbine Blades

Design Perspective

- A good Helicopter rotor airfoil
  ≠ good Wind Turbine rotor airfoil

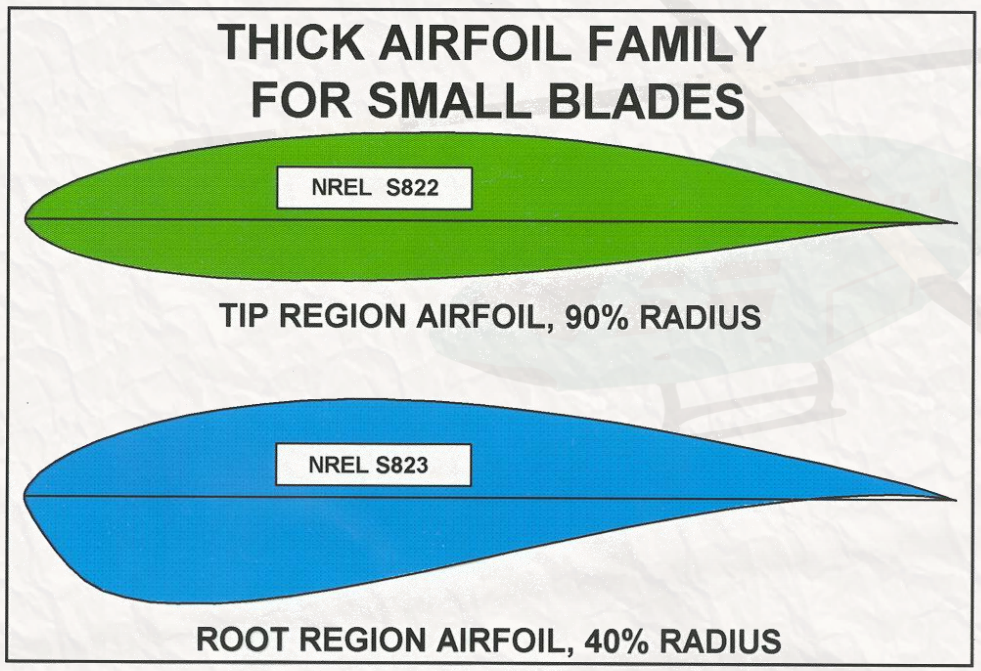
- Operates in a very different environment
  - Airborne particles get stuck to the blades causing roughness effects (Difficult to clean blades regularly)
  - Huge structural and aerodynamic loading (in a 1.5 MW turbine, each blade weigh ~ 5 tons)



Design Philosophy and Criteria

- Specially-tailored airfoils for wind turbines
- Large thickness airfoils at the root. Thickness reduced closer to the tip
- Criteria
  - Moderate to high thickness ratio t/c
    - Rigid rotor: 16%–26% t/c
    - Flexible rotor: 11%–21% t/c
    - Small wind turbines: 10%-16% t/c
  - High lift-to-drag ratio
  - Minimal roughness sensitivity
  - Weak laminar separation bubbles

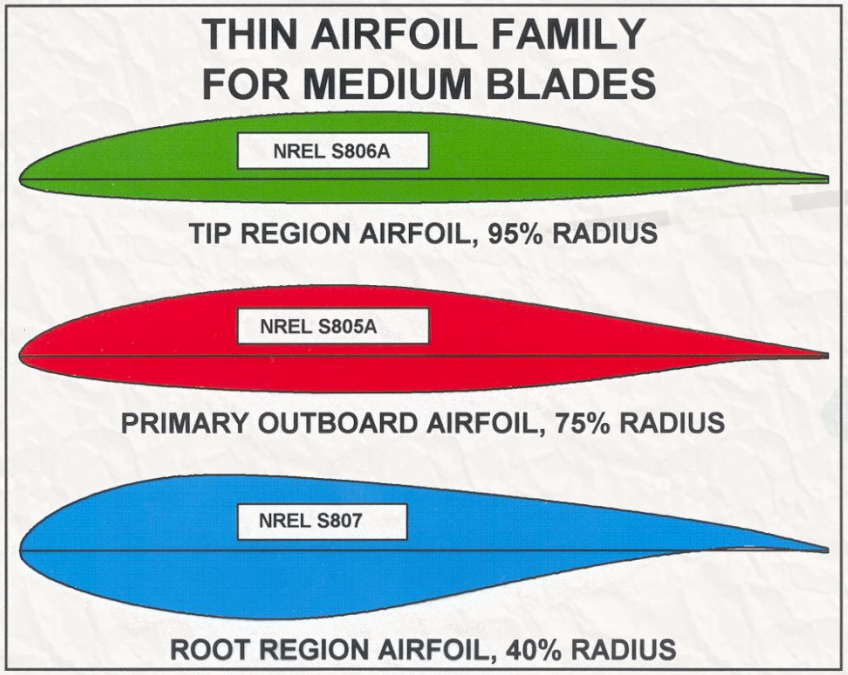
Airfoils Based on Rotor Sizes



Design Parameters

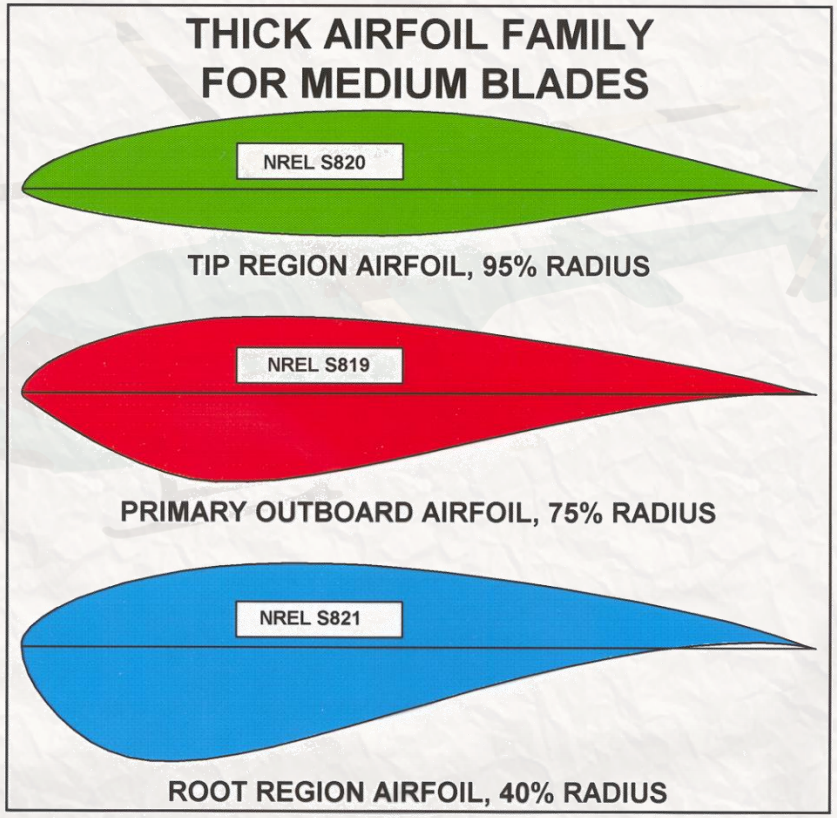
Airfoil	r/R	Re. No. ($x10^6$)	t/c	C_{lmax}	C_{dmin}	C_{mo}
S822	0.90	0.6	0.160	1.00	0.010	-0.07
S823	0.40	0.4	0.240	1.30	0.018	-0.15

Airfoils Based on Rotor Sizes



Design Parameters

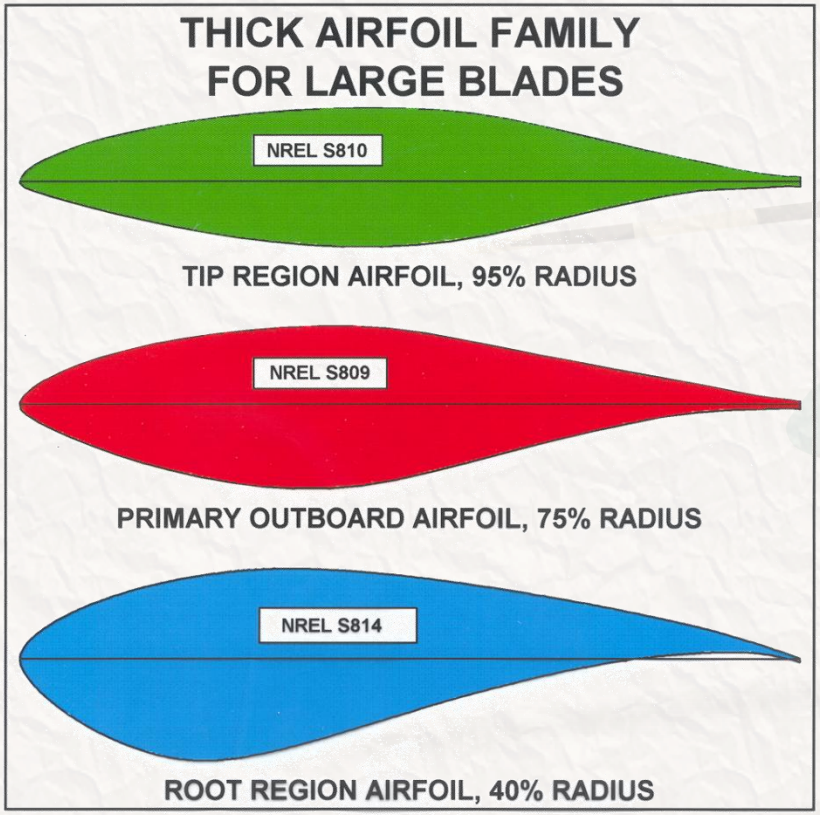
Airfoil	r/R	Re. No. ($x10^6$)	t/c	$C_{l_{max}}$	$C_{d_{min}}$	C_{m_o}
S806A	0.95	1.3	0.115	1.10	0.004	-0.05
S805A	0.75	1.0	0.135	1.20	0.005	-0.05
S807	0.40	0.8	0.180	1.46	0.010	-0.10
S808	0.30	0.8	0.210	1.30	0.012	-0.12

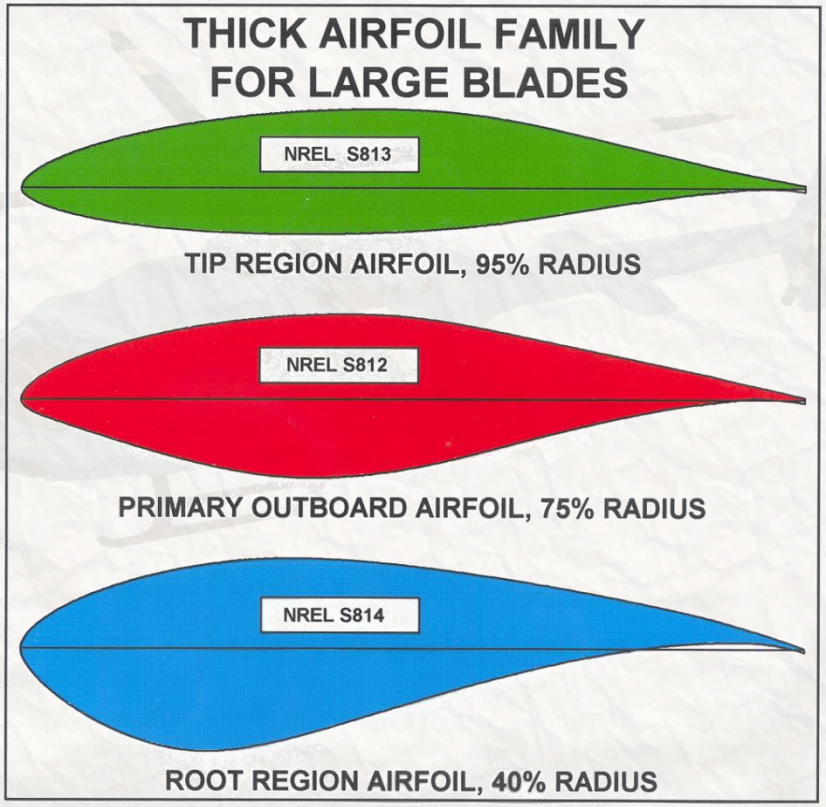


Design Parameters

Airfoil	r/R	Re. No. ($x10^6$)	t/c	$C_{l_{max}}$	$C_{d_{min}}$	C_{m_o}
S820	0.95	1.3	0.160	1.10	0.007	-0.07
S819	0.75	1.0	0.210	1.20	0.008	-0.07
S821	0.40	0.8	0.240	1.40	0.014	-0.15

Airfoils Based on Rotor Sizes





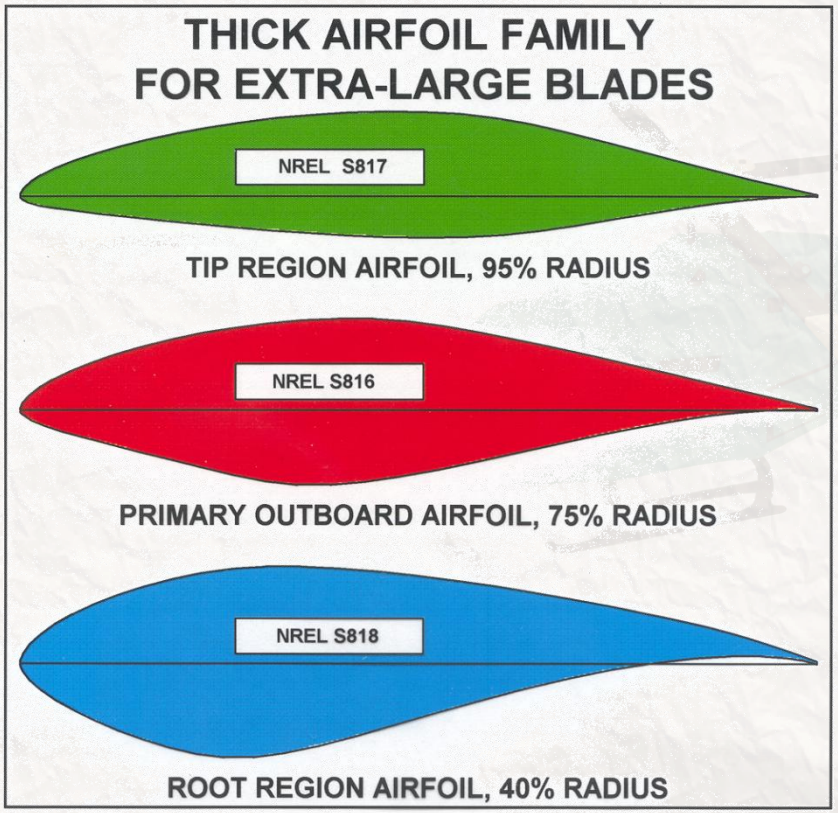
Design Parameters

Airfoil	r/R	Re. No. ($x10^6$)	t/c	C_{lmax}	C_{dmin}	C_{mo}
S810	0.95	2.0	0.180	0.90	0.006	-0.05
S809	0.75	2.0	0.210	1.00	0.007	-0.05
S814	0.40	1.5	0.240	1.30	0.012	-0.15
S815	0.30	1.2	0.260	1.10	0.014	-0.15

Design Parameters

Airfoil	r/R	Re. No. ($x10^6$)	t/c	C_{lmax}	C_{dmin}	C_{mo}
S813	0.95	2.0	0.160	1.10	0.007	-0.07
S812	0.75	2.0	0.210	1.20	0.008	-0.07
S814	0.40	1.5	0.240	1.30	0.012	-0.15
S815	0.30	1.2	0.260	1.10	0.014	-0.15

Airfoils Based on Rotor Sizes



Design Parameters

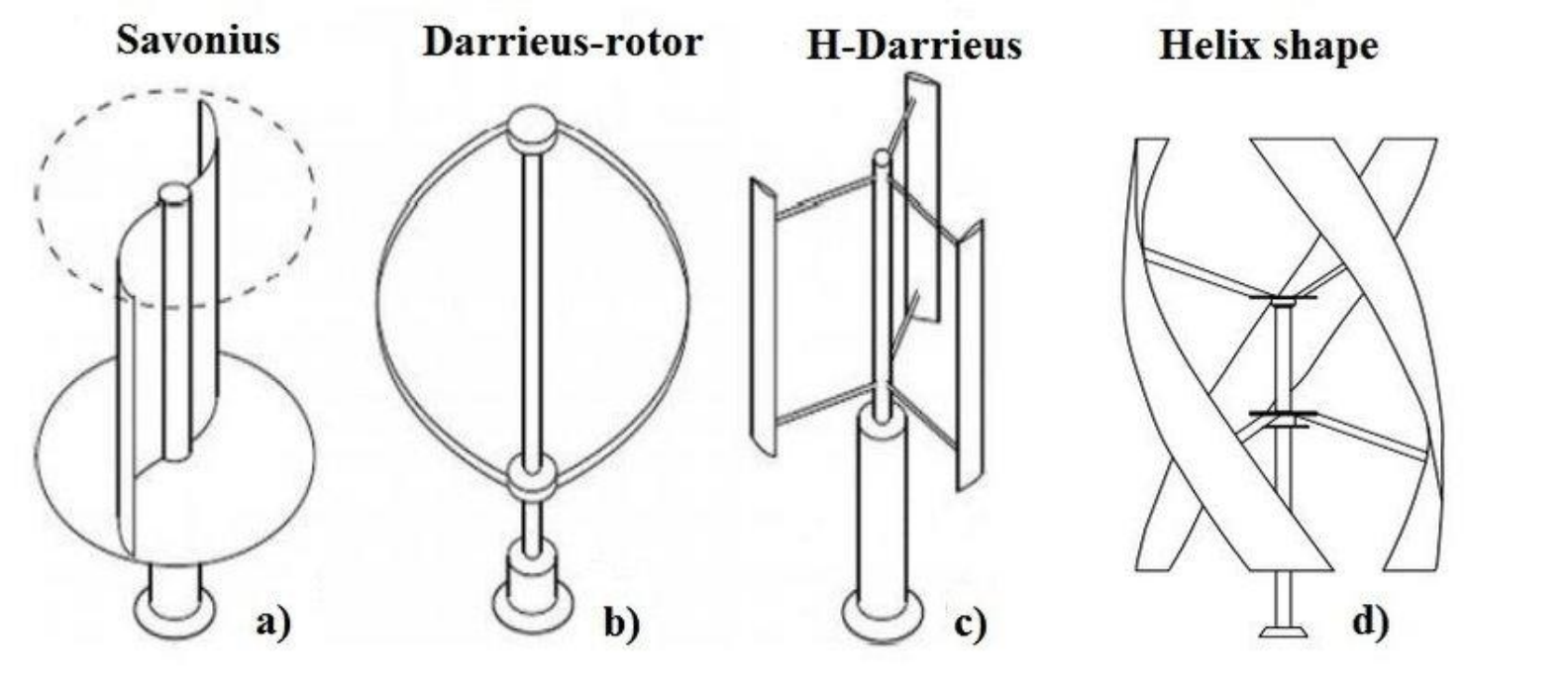
Airfoil	r/R	Re. No. ($x10^6$)	t/c	C_{lmax}	C_{dmin}	C_{mo}
S817	0.95	3.0	0.160	1.10	0.007	-0.07
S816	0.75	4.0	0.210	1.20	0.008	-0.07
S818	0.40	2.5	0.240	1.30	0.012	-0.15

Airfoil Selection

- Appropriate design Reynolds number
- Airfoil thickness: Centrifugal stiffening and desired blade rigidity
- Roughness Insensitivity

Vertical Axis Wind Turbines

Vertical Axis Wind Turbines (VAWT)



Castellani, Francesco & Astolfi, Davide & Peppoloni, Mauro & Natili, Francesco & Buttà, Daniele & Hirschl, Alexander. (2019). Experimental Vibration Analysis of a Small Scale Vertical Wind Energy System for Residential Use. Machines. 7. 35. 10.3390/machines7020035.

VAWT vs HAWT (Efficiency)

VAWT

- Blades see extreme AoA. Stall and Deep Stall
- Lower Efficiency
- Efficiency improves with rotor speed

HAWT

- Blades designed for optimal AoA of airfoils
- Higher Efficiency





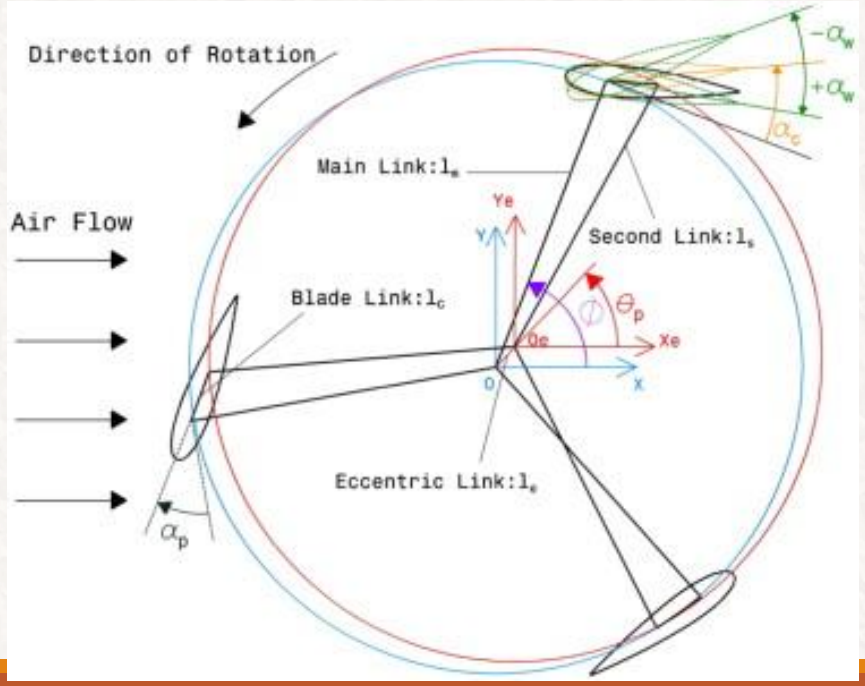
VAWT vs HAWT (Low Speed Performance)

VAWT

- More stall at low speeds
- Difficulty starting
- Need external starting mechanisms

HAWT

- Rotor blade pitch angle adjusted for low speeds
- Self starting



Elkhoury, M., Kiwata, T., & Aoun, E. (2015). Experimental and numerical investigation of a three-dimensional vertical-axis wind turbine with variable-pitch. *Journal of wind engineering and Industrial aerodynamics*, *139*, 111-123.

VAWT vs HAWT (Wind Direction)

VAWT

- Insensitive to wind direction
- No need of yawing mechanism
- Can capture energy from gusty wind

HAWT

- Highly sensitive to wind direction
- Yawing mechanism is must
- Does not respond fast enough to gusts



VAWT vs HAWT (Manufacture, Assembly, Cost)

VAWT

- Untwisted, Rectangular blades
- Gear box and generator on ground
- Cheap to build and maintain

HAWT

- Huge, Twisted, Tapered blades
- Gear box and generator on tower
- Expensive to build and maintain



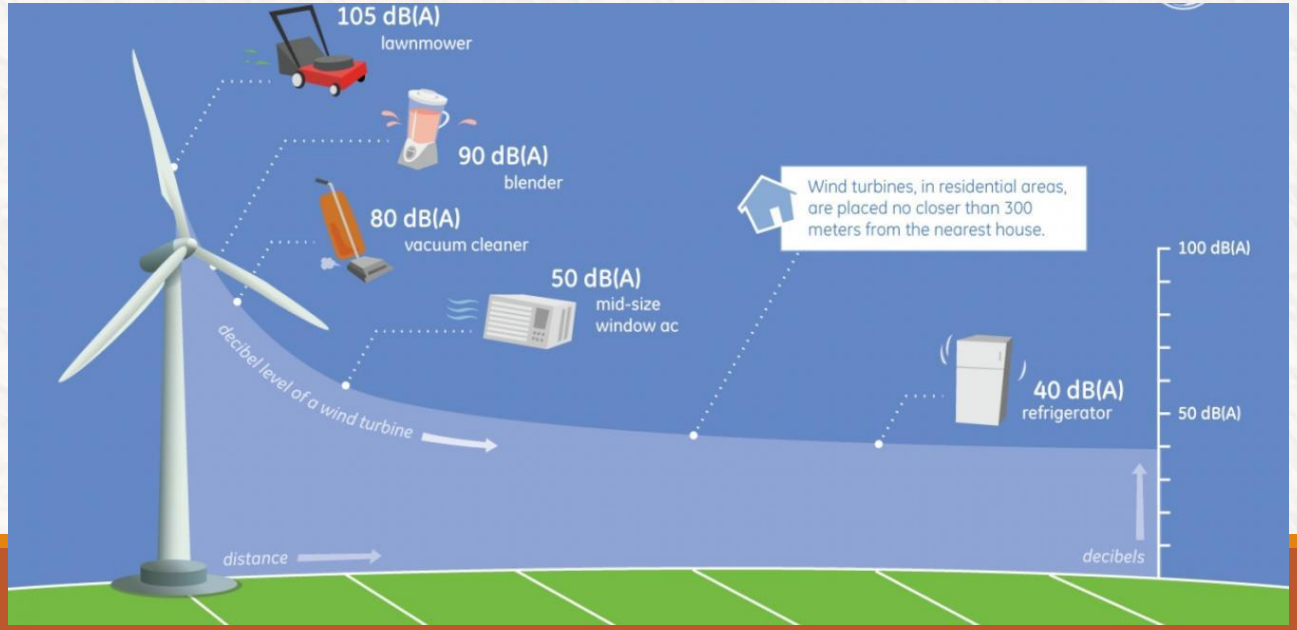
VAWT vs HAWT (Noise & Environmental)

VAWT

- Low tip speed (small R)
- Gear box near ground
- Low aerodynamic & mechanical noise
- No avian issues

HAWT

- High tip speed (large R)
- Gearbox on a tower
- High aerodynamic and far reaching mechanical noise
- Bird strike and death problem



VAWT vs HAWT (Installation)

VAWT

- Can be installed in open fields as well as on roof tops



HAWT

- Open field necessary



Ship / Aquatic Propellers

Aircraft vs Ship Propellers









Aircraft vs Ship Propellers

Governing Factors

- Fluid properties: Density and viscosity
- Vehicle Speed
- Cavitation
- Weight



Ship Propeller Design Aspects

Propeller Diameter

- Larger, the better! : Momentum Theory
- But… Structural & Manufacturing issues



Emma Maersk: 131 ton weight propeller. Largest in the world (German)

Ship Propeller Design Aspects

Pitch (Fixed vs Variable)

- Variable: Manoeuvrability, Efficiency at higher complexity, cost
- Fixed: Reliable, Efficient only at certain speeds



Fixed Variable

Ship Propeller Design Aspects

Propeller Speed

- Resonance: Structural fatigue
- Cavitation: Noise, Blade Erosion

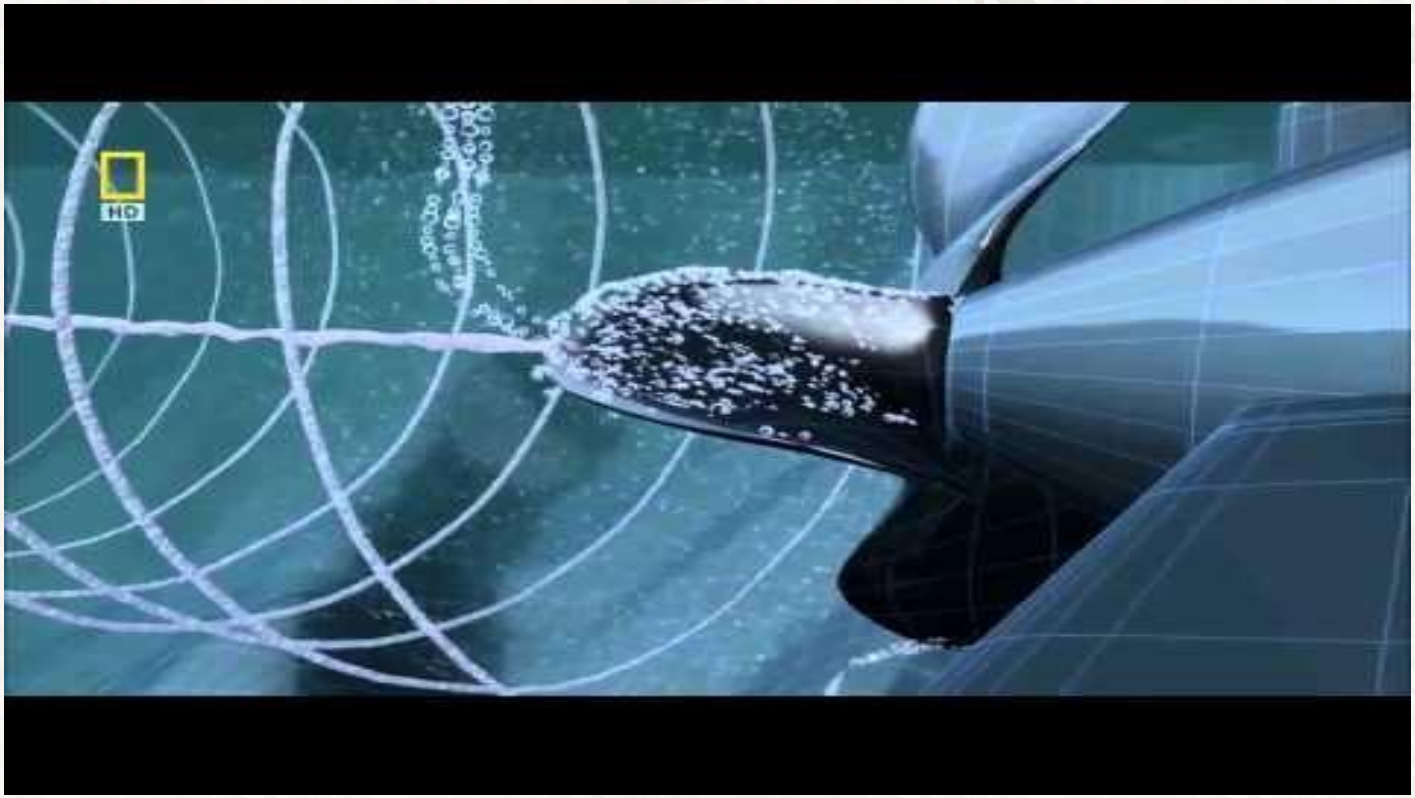




Ship Propeller Design Aspects

Propeller Speed

- High Speed: Cavitation, Noise



https://youtu.be/ON_irzFAU9c

Ship Propeller Design Aspects

Solidity

- Very different from Aircraft Propellers
- Why?
- High density of water & Low speed of vehicle

$$v = -\frac{V}{2} + \sqrt{\left(\frac{V}{2}\right)^2 + \frac{T}{2\rho A}}$$

➔ Thrust possible at induced velocities

➔ Low RPM requirement

- Low speed, but high thrust force

➔ More structural strength needed

- Also, inertial effects more dominant than viscous effects

	Water	Air	
Density	997	1.225	kg/m3
Viscosity	0.0010016	0.0000181	Pa-s
ρ/μ	9,95,407.35	67,679.56	

15 times greater Reynolds number for the same dimensions and speed!

Ship Propeller Design Aspects

Shape & Number of Blades

- Very curved leading edge: Smooth propulsion
- High number of blades: Reducing unsteady forces, Higher efficiency

